2.4 Algebraic Reasoning

Name _____ Period ____ Date _____

Geometry 2.4 · Student sheet

I can…

- justify each step when solving an equation.
- solve a formula for a variable and justify the steps.
- use the Reflexive, Symmetric, and Transitive Properties.

Vocabulary · fill in the missing word

Reflexive Property — a = _____

Symmetric Property — if a = _____, then b = a

Transitive Property — if a = _____ = c, then a = c

Part 1 · Discovery: name your moves

Solve 3x + 8 = 23. Next to each step, write what you did in your own words.

Step	What I did	Property name (from the class)
3x + 8 = 23	Given	—

Part 2 · Notes and examples (fill in each note, then work the example)

Properties of equality: Addition, Subtraction, Multiplication, and Division: do the same thing to _____.

Example 1

Solve 3x + 8 = 23. Justify each step.

Distributive Property: a(b + c) = _____. **Substitution Property:** if a = b, then a can replace _____ in any equation or expression.

Example 2

Solve −2(x + 3) = 4x + 18. Justify each step.

Example 3

Solve $A = \frac{1}{2}bh$ for h. Justify each step.

Reflexive, Symmetric, and Transitive also apply to segment lengths and angle _____.

Example 4

Name the property.

a. $m\angle A = m\angle A$

b. If $AB = CD$, then $CD = AB$.

c. If $m\angle 1 = m\angle 2$ and $m\angle 2 = m\angle 3$, then $m\angle 1 = m\angle 3$.

Example 5

B is between A and C. $AC = 32$, $AB = 3x + 2$, and $BC = 5x - 2$. Solve for x with a reason for each step.

Part 3 · Partner practice

Partner 1

Solve $5x - 7 = 2x + 11$. Justify each step.

Partner 2

Name the property: If $JK = KL$ and $KL = 12$, then $JK = 12$.